

SOUTH ST PAUL, MN, USA

WMO: 726603

Lat: 44.857N

Lon: 93.033W

Elev: 250

StdP: 98.36

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04974

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-20.3	-27.5	0.3	-21.9	-25.7	0.4	-19.9	9.2	-9.3	8.3	-8.6	1.9	280	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	32.5	22.7	31.0	21.9	29.1	20.9	24.9	30.0	23.7	28.5	22.6	27.4	4.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.3	18.6	27.7	22.2	17.4	26.7	21.1	16.2	25.5	77.0	29.8	71.9	28.7	67.7	27.3	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.5	6.7	-26.2	35.2	3.5	1.9	-28.7	36.5	-30.8	37.7	-32.8	38.7	-35.3	40.1		
			-26.4	26.6	3.4	1.6	-28.8	27.7	-30.8	28.6	-32.6	29.5	-35.1	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-8.3	-6.1	0.6	8.3	15.1	20.4	23.3	21.8	17.7	9.8	1.9	-5.3
	DBStd	12.22	6.82	6.46	6.59	5.56	4.65	3.74	2.99	2.95	4.55	4.84	5.72	6.24
	HDD10.0	2214	568	452	298	95	9	0	0	0	3	64	250	476
	HDD18.3	4077	826	685	549	303	123	21	2	4	66	269	494	734
	CDD10.0	1604	0	0	7	44	167	313	412	366	233	57	6	0
	CDD18.3	427	0	0	1	2	22	84	156	112	47	4	0	0
	CDH23.3	3632	0	0	2	28	232	700	1410	847	373	41	0	0
	CDH26.7	1138	0	0	0	7	72	217	494	237	104	7	0	0
Wind	WSAvg	2.7	2.7	2.8	2.9	3.3	2.9	2.5	2.4	2.2	2.4	2.8	2.8	2.7
Precipitation	PrecAvg	841	21	18	44	75	106	125	114	116	84	69	44	27
	PrecMax	1038	59	39	103	160	221	236	284	222	169	143	133	74
	PrecMin	607	3	3	9	23	22	55	49	29	16	18	2	6
	PrecStd	111	13	9	23	33	48	48	54	52	49	39	35	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	10.9	21.4	27.2	32.3	33.8	35.0	33.7	32.3	27.7	19.5	10.0
		MCWB	3.7	7.4	14.7	15.9	19.9	22.9	24.5	22.8	21.7	17.8	13.0	7.8
	2%	DB	4.1	7.1	17.0	22.5	28.4	31.2	32.7	31.2	29.4	23.7	16.0	6.8
		MCWB	1.9	4.1	12.1	13.0	17.8	21.5	23.6	22.4	20.8	16.1	11.1	4.3
	5%	DB	2.4	4.1	13.0	20.0	26.1	29.0	31.2	29.0	27.5	20.7	12.8	4.1
		MCWB	1.1	2.1	8.7	11.6	16.9	20.3	22.6	21.5	19.8	14.8	8.7	2.2
	10%	DB	1.0	2.6	9.8	17.5	23.0	27.4	29.2	27.6	25.4	17.6	10.1	2.6
		MCWB	-0.1	0.9	6.5	10.1	15.2	19.1	21.4	20.4	18.7	12.7	6.8	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	7.5	16.0	16.7	22.5	25.0	26.9	25.7	24.0	20.2	14.5	8.7
		MCDB	7.0	10.3	19.2	24.9	28.9	30.8	31.5	30.9	29.1	23.9	18.0	9.6
	2%	WB	2.5	4.3	12.4	14.5	19.9	23.1	25.4	24.3	22.5	17.3	11.7	4.4
		MCDB	4.0	6.3	16.2	20.3	25.3	28.9	30.7	28.5	26.9	21.4	15.1	6.3
	5%	WB	1.2	2.6	9.3	12.7	18.3	21.9	24.0	23.2	21.2	15.2	9.0	2.6
		MCDB	2.4	4.2	13.0	18.3	23.4	27.0	29.0	27.2	25.4	19.0	12.3	4.0
	10%	WB	-0.1	1.0	6.5	11.0	16.7	20.6	22.8	22.0	19.9	13.5	7.1	1.3
		MCDB	0.9	2.5	9.5	16.6	22.0	25.3	28.0	25.8	24.3	17.3	9.8	2.6
Mean Daily Temperature Range		MDBR	8.0	8.4	8.8	10.1	10.4	9.7	9.9	9.6	10.2	9.4	7.9	7.1
	5% DB	MCDBR	9.0	9.5	13.1	14.5	14.1	12.1	11.5	11.0	12.4	14.2	12.1	9.1
		MCWBR	7.4	7.4	8.3	7.6	6.0	5.2	4.8	4.7	5.5	7.6	7.8	7.4
	5% WB	MCDBR	8.8	8.9	12.3	12.4	12.1	10.3	10.7	9.5	10.7	11.3	11.5	8.4
		MCWBR	7.4	7.2	8.3	7.6	6.2	5.6	5.5	5.5	5.7	7.2	8.0	7.0
Clear-Sky Solar Irradiance	taub	0.268	0.289	0.324	0.378	0.408	0.402	0.400	0.405	0.377	0.324	0.292	0.270	
	taud	2.418	2.341	2.371	2.289	2.257	2.320	2.346	2.334	2.378	2.517	2.512	2.437	
	Ebn at Noon	862	900	911	882	863	868	865	846	842	853	827	820	
	Edh at Noon	71	93	105	124	133	125	121	118	104	79	66	64	
All-Sky Solar Radiation	RadAvg	1.73	2.81	3.79	4.59	5.32	5.93	6.29	5.25	4.10	2.61	1.67	1.21	
	RadStd	0.20	0.29	0.35	0.47	0.43	0.47	0.23	0.37	0.29	0.35	0.16	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page